Attachment 1 - SOP-463

Pump type code	Pump speed [rpm]	Head for stage [m]	Power per stage [kW]	Shaft diameter [mm]	Fitting type	Tolerance grade	Max number of stages on same shaft diameter	Modification required to previous design	Tolerance grade TMP	Tolerance grade GAB		
OH1	--	--	--	--	Push fit	H7/h6	--	--	H7/h6	--		
OH2	--	--	--	--	Push fit	H7/h6	--	--	H7/h6	H7/h6		
OH3	--	--	--	--	Push fit	H7/h6	--	--	--	H7/h6		
OH5	--	--	--	--	Push fit	H7/h6	--	--	--	H7/h6		
OH6	--	--	--	--	--	--	--	--	*	*		
BB1	≤ 3800	≤ 200	≤1500	≤140	Push fit	H7/h6	2	--	H7/h6	H7/h6	no pompe GAB sopra i 1500kW/stadio	
				>140	Push fit	Double cylindrical fitting H7 / h6	1	yes				
		--	--	≤140	Push fit	H6/h5	1	yes	H6/h5	H7/h6		
		--	--	>140	Push fit	Double cylindrical fitting H6 / h5	1	yes	no standard (vedere foglio aggiuntivo)	---		
	> 3800	--	--	--	Shrink fit	H6/p6	1	yes	--	--		
BB2	≤ 3800	≤ 200	≤1500	≤140	Push fit	H7/h6	2	--	H7/h6	H7/h6	no pompe GAB sopra i 1500kW/stadio	
				>140	Push fit	Double cylindrical fitting H7 / h6	1	yes				
		--	--	≤140	Push fit	H6/h5	1	yes	H6/h5	H7/h6		
		--	--	>140	Push fit	Double cylindrical fitting H6 / h5	1	yes	H6/h5 min 0.1 mm	H7/h6		
	> 3800	--	--	--	Shrink fit	H6/p6	1	yes	--	P6/h6		
BB3	≤ 3800	≤ 200	--	--	Push fit	H6/h6	1	To be defined at internal KOM	H6/h5	Push H6/h6	not more then 3 impellers with same nominal bore (TMP)	
		> 200	--	--	Shrink fit	H6/p5 IL P5/h6 B/B	2	yes	H6/h5 min 0.1 mm	Push H6/h6		
	> 3800	--	--	--	Shrink fit	H6/p5 IL P5/h6 B/B	2	yes	H6/h5 min 0.1 mm	Light shrink fit P6/h6	not more then 3 impellers with same nominal bore (TMP)	GAB uguale BB3/BB5
BB4/BB5	≤ 3800	≤ 200	--	--	Push fit	H7/h6	1	To be defined at internal KOM	H6/h5	Push H6/h6		
		> 200	--	--	Shrink fit with impeller axally limited in both	H6/p5 IL P5/h6 B/B	2	yes	H6/h5 min 0.1 mm	Push H6/h6		
			--	--	Shrink fit with impeller axally limited in one	H6/r5 IL R6/h5 B/B	2	yes	H6/h5 min 0.1 mm	Push H6/h6		
	> 3800	--	--	--	Shrink fit with impeller axally limited in both	H6/p5 IL P5/h6 B/B	2	yes	H6/h5 min 0.1 mm	Light shrink fitP6/h6		
		--	--	--	Shrink fit with impeller axally limited in one	H6/r5 IL R6/h5 B/B	2	yes	H6/h5 min 0.1 mm	Light shrink fit P6/h6		
VS1/VS2/VS6/VS7	≤ 1800	--	--	<90	Push fit	H7/g6	4	To be defined at internal KOM	H7/g6	H7/h6	not more then 4 impellers with same nominal bore (GAB) not more then 3 impellers with same nominal bore (GAB)	
		--	--	90<D<140	Push fit	H7/g6	3	yes	H7/g6	H7/h6		
	≤ 3600	--	--	< 90	Push fit	H6/g6	4	To be defined at internal KOM	H7/g6	H7/h6		
		--	--	90<D<140	Push fit	H6/g6	3	yes	H6/g6	H7/h6		
	≤ 1800	--	--	> 140	Push fit	Tapered 1:16 (single stage)	--	yes	Tapered 1:16 (single stage)	--		
			--		Push fit	Double cylindrical fitting (multistage) H7 / g6	1	yes	Double cylindrical fitting (multistage) H7 / g6	--		
			--		Push fit	Tapered 1:16 (single stage)	--	yes	Tapered 1:16 (single stage)	--		

	≤ 3600	--	--	> 140	Push fit	Double cylindrical fitting (multistage) H6 / g6	1	yes	Double cylindrical fitting (multistage) H6 / g6	--
VS4	--	--	--	--	Push fit	H7/h6	--	yes	H7/h6	H7/h6
Concrete volute vertical pump	--	--	--	--	Push fit	Tapered 1:16	--	yes	Tapered 1:16	--

IL In line

B/B back to back

ipment Ye	Internal Order	Job	Customer	Engineering	End User	Country of installation	Plant Location	Plant	ITEM	Pump Service	Pump Model	API Pump Type	Api Material class	Capacity(m3/h)	Head (m)	Pump speed	Driver Power(kW)	Rotor fit	Coating on shaft	Pump Size	stadi	
2022	T30704001	30704	HORSOL SWITZ Engineering Asia Pte L		PUB - PUBLIC UTILITIES BOARD SINGAPORE	Singapore		PUB PROJECT C3B1 650MLD CERAMIC WWTP		SEA WATER SUPPLY		BB1	D2	6609	72	984	1750	130 H7/h6	No	750DD80	1	0,7
2021	T30691002	30691	ACCIONA AGUA S.A.U.		SWCC	Saudi Arabia	Al Khobar	AL KHOBAR SWRO PH II		BOOSTER TO ERD		BB1	D2	13346	43,5	595	1900	205 H7/h6	No	1000DD120	1	0,3
2021	T30691005	30691	ACCIONA AGUA S.A.U.		SWCC	Saudi Arabia	Al Khobar	AL KHOBAR SWRO PH II		BOOSTER TO HP		BB1	D2	8167	134,4	980	3650	180 H7/h6	No	700DD100	1	1,3
2020	T30660001	30660	SK ecoplant		ADNOC ONSHORE - ADCO	United Arab Emirates	Fujairah	MANDOUS		UNLOADING PUMP		BB1	S6	4240	138,5	1485	1850	133.1 H6/p6	Yes	450DD75	1	2,1
2018	T30638002	30638	ALRASHID TRADING AND		SWCC	Saudi Arabia	Shoaibah	SHOAIBA-MINA LINE B Shoaiba Extension		MAIN PUMPS		BB1	NONE	6480	289	1782	6800	188 H6 / 188.07 h5	No	750DD85	1	5,1
2018	T30638004	30638	ALRASHID TRADING AND		SWCC	Saudi Arabia	Shoaibah	SHOAIBA-MINA LINE B Shoaiba Extension		MAIN PUMPS		BB1	NONE	5064,5	552	1782	10200	211 H5 / 211 h5	Yes	600D290	2	2,5
2016	T30612001	30612	SIEMENS DD -ZAGREB		JANAF DD	Croatia		OMISALI TANKER LOADING SYSTEM		CRUDE OIL SHIPPING		BB1		5000	95	743	1600	190 H7 /190 h6	no	600DD115	1	0,7
2016	T30598001	30598	PIPELINE ENGINEERING L.L.P. - LONDO		TRANSNEFT	Russian Federation		ESPO PROJECT 1 & 2		MAIN OIL PUMP		BB1		9591	376	2992	12000	198H7/js6	no	500DD60	1	11,2
2016	T30598002	30598	PIPELINE ENGINEERING L.L.P. - LONDO		TRANSNEFT	Russian Federation		ESPO PROJECT 1 & 2		MAIN OIL		BB1		6149	184	3000	6300	H7/js6	no	450DD75	1	5,5
2015	T30605003	30605	Transneft Oil Pumps JSC		TRANSNEFT	Russian Federation		TRANSNEFT		MAIN OIL		BB1	NONE	9400	210	3000	6600	H7/js6	no	500DD50	1	6,3
2015	T30605004	30605	Transneft Oil Pumps JSC		TRANSNEFT	Russian Federation		TRANSNEFT		MAIN OIL		BB1	NONE	7000	210	3000	6300	H7/js6	no	450DD50	1	6,3
2015	T30605005	30605	Transneft Oil Pumps JSC		TRANSNEFT	Russian Federation		TRANSNEFT		MAIN OIL		BB1	NONE	3600	230	3000	2600	H7/js6	no	400DD45	1	6,9
2015	T30605006	30605	Transneft Oil Pumps JSC		TRANSNEFT	Russian Federation		TRANSNEFT		MAIN OIL		BB1	NONE	4880	143	3000	2600	H7/js6	no	400DD45	1	4,3
2015	T30605007	30605	Transneft Oil Pumps JSC		TRANSNEFT	Russian Federation		TRANSNEFT		MAIN OIL		BB1	NONE	3600	230	3000	2600	H7/js6	no	400DD45	1	6,9
2015	T30605008	30605	Transneft Oil Pumps JSC		TRANSNEFT	Russian Federation		TRANSNEFT		MAIN OIL		BB1	NONE	7000	210	3000	6600	H7/js6	no	500DD50	1	6,3
2015	T30605010	30605	Transneft Oil Pumps JSC		TRANSNEFT	Russian Federation		TRANSNEFT		MAIN OIL		BB1	NONE	7000	210	3000	6600	H7/js6	no	500DD50	1	6,3
2015	T30586002	30586	BRUNO PRESEZZI SPA		PARDIS PETROCHEMICAL	Iran		PARDIS PETROCHEMICAL - ASALUYEH		COOLING CIRCUIT		BB1	S6	20000	60,5	594	4120	240H7/g6	no	1200DD135	1	0,4
2015	T30574003	30574	LARSEN & TOUBRO LIM. - VADODARA - I		NUCLEAR POWER CORPORATION OF INDIA LTD	India		RAJASTAN ATOMIC POWER PROJ 7&8		BOOSTER		BB1		2175	98,6	1485	4100			300DDH56	1	1,5
2014	T30550001	30550	AL RASHID TRADING & CONTRACTING CO.		NWC - NATIONAL WATER COMPANY	Saudi Arabia	Riyadh	SAAD		WATER TRANSMISSION		BB1		3654	244	1188	3500	175H7/h6	no	450D2D75	2	0,7
2013	T53093001	53093	SALINE WATER CONVERSION CORP.		SWCC	Saudi Arabia	Shoaibah	SHOAIBAH-JEDDAH WATER TRANSMISSION		WATER TRANSMISSION		BB1		1894	400	3550	3650			300D2D45	2	3,6
2012	T30535001	30535	NEYRPERSE		ILC-IRAN LIQUEFIED NATURAL GAS CO.	Iran		KANGAN COMBINED CYCLE POWER PLANT		FRESH WATER		BB1		17169	35	425	2050	220H7/g6	no	1200DD135	1	0,1
2011	T30512002	30512	SAUDI SERVICE FOR ELECTRO MECHANIC		SWCC	Saudi Arabia	Shoaibah	SHOAIBA - MINA PH 3 - WATER TRANSMISSION		MAIN LINE		BB1		5065	559	1782	10280			500D2D80	2	2,5
2009	T30474002	30474	CONSORTIUM OF YUKSEL INSAAT A.S..YU		SWCC	Saudi Arabia	Khobar	EASTERN PROVINCE PH II		MAIN LINE		BB1		7312	310	1782	8126	doppio diametro su due fasce del circa 10% della lunghezza della girante 194/194,5 - H6/h6	no	700DD90	1	5,5
2007	T30435002	30435	SAUDI SERVICE FOR ELECTRO MECHANIC		SWCC	Saudi Arabia	Shoaibah	SHOAIBA PHASE 3 - WATER TRANSMISSION		MAIN LINE		BB1		3988	216	3564	3000	95H7/h6	No	400DD40	1	7,7
2007	T30435003	30435	SAUDI SERVICE FOR ELECTRO MECHANIC		SWCC	Saudi Arabia	Shoaibah	SHOAIBA PHASE 3 - WATER TRANSMISSION		MAIN LINE		BB1		6480	116	1782	2700	125H7/h6	No	600DD54	1	2,1
2007	T30435004	30435	SAUDI SERVICE FOR ELECTRO MECHANIC		SWCC	Saudi Arabia	Shoaibah	SHOAIBA PHASE 3 - WATER TRANSMISSION		MAIN LINE		BB1		5065	552	1782	9650	182H5/h6	Si	500D2D80	2	2,5